COGNIEASE

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PROBLEM STATEMENT

People with ADHD or fatigue often find digital interfaces harder to use when mentally tired.

INTERFACE A BARRIER?

For ADHD and Chronic Fatigue Users, Everyday Interfaces Become Barriers.

Our digital world is a constant series of interfaces:

- Work & Study
- Banking & Healthcare
- Communication
- Everything runs through screens

People having issues with interfaces face a significant disadvantage in their professional lives.

FocusView: Understanding and Customizing Informational Video Watching Experiences for Viewers with ADHD

Hanxiu 'Hazel' Zhu, Ruijia Chen, Yuhang Zhao

While videos have become increasingly prevalent in delivering information across different educational and professional contexts, individuals with ADHD often face attention challenges when watching informational videos due to the dynamic, multimodal, yet potentially distracting video elements. To understand and address this critical challenge, we designed \textit{FocusView}, a video customization interface that allows viewers with ADHD to customize informational videos from different aspects. We evaluated FocusView with 12 participants with ADHD and found that FocusView significantly improved the viewability of videos by reducing distractions. Through the study, we uncovered participants' diverse perceptions of video distractions (e.g., background music as a distraction vs. stimulation boost) and their customization preferences, highlighting unique ADHD-relevant needs in designing video customization interfaces (e.g., reducing the number of options to avoid distraction caused by customization itself). We further derived design considerations for future video customization systems for the ADHD community.

Subjects: **Human-Computer Interaction (cs.HC)**
Cite as: arXiv:2507.13309 **[cs.HC]**

SOLUTION

An adaptive AI system that passively detects cognitive fatigue through user behavior (typing, scrolling, cursor movement) and simplifies the interface —reducing complexity and improving usability.

The TECH *Solution*

01 Data Collection

The Chrome Extension monitors behavioral cues like typing speed, error rates, cursor jitter, and idle time.

02 AI Processing

Captured input is sent to a custom-trained AI, which calculates a fatigue level from 1 to 5.

03 Interface Adaptation

The extension simplifies the UI in real-time based on the user's current cognitive load.

04 Outcome

Adaptive simplification helps boost productivity and maintain focus throughout the day.

Why our solution is unique?

REAL TIME behaviour detection

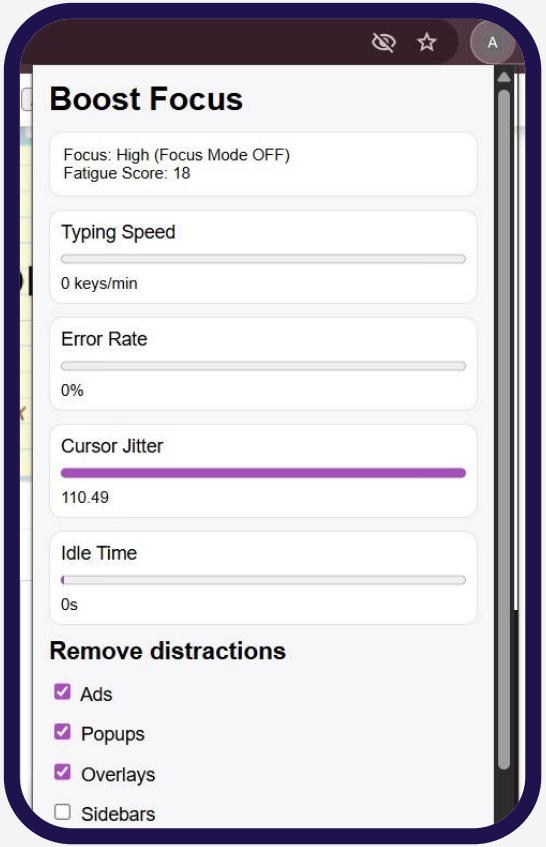
- Our system detecs passive signals of the user in real time in the background
- Detects anywhere on web
- Detects multiple parameters for high accuracy.
- Typing speed, jitter speed, error rate, idle time

CUSTOM TRAINED AI

- We trained our ai to work on 4 parameters taking input from our chrome extension
- Predicts fatigue level with high accuracy(trained on our made dataset)

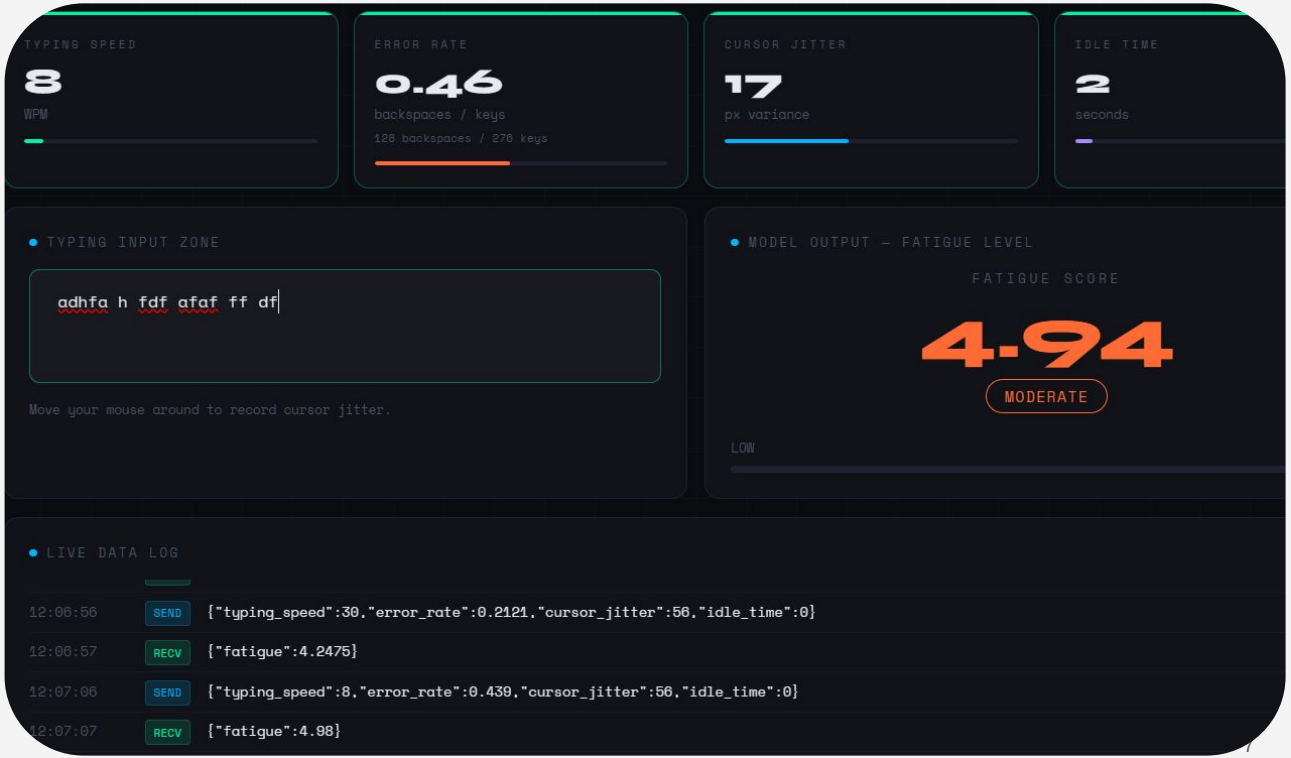
Interface simplification ACROSS THE WEB

- Simplifies almost any website on internet, hiding unnecessary clutter, ads etc.
- Increases font size, and reduced overall complexity of the webpage

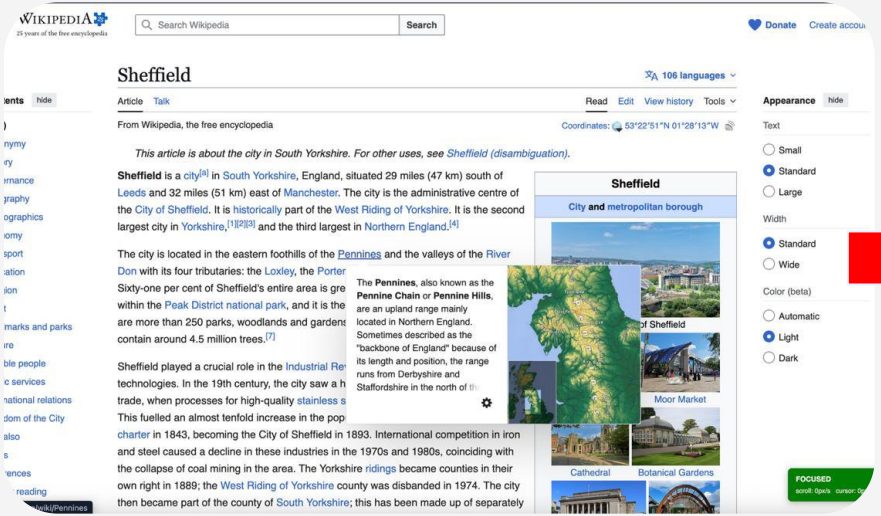


CUSTOM TRAINED AI MODEL

We trained a custom AI model that passively analyzes typing rhythm, cursor jitter, scroll behavior, idle gaps, and error patterns to estimate the user's real-time fatigue level — enabling the interface to adapt before performance collapses.



INTERFACE SIMPLIFICATION



Thank you